QoE-Aware Deep Reinforcement Learning for URLLC-Prioritized Slice Admission Control: From Offline Simulation to a Live O-RAN Testbed

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***Abstract*—Deep reinforcement learning (DRL) is widely proposed for O-RAN slice admission control, but the majority of this work – including our own [1], [2] – is validated only in simulation, a gap our systematic review of 67 core 2024–2026 studies formalized as the field's *validated-deployment gap* [3]. This paper closes that gap and centers the evaluation on quality of experience (QoE): we deploy a QoE-aware DQN admission policy ($r = \alpha \cdot$ MOS $- \beta \cdot$ cost $- \gamma \cdot$ SLA_viol) on a real OpenAirInterface (OAI) gNB across three slices (eMBB, URLLC, mMTC), compare it against a static baseline and its SLA-only counterpart, and report both live and offline-simulated results side by side. Live, DQN holds $100.0 \pm 0.0\%$ SLA compliance against the baseline's seed-dependent $73.6 \pm 18.6\%$ (worst episode 0.0%) across 45 episodes. A held-out, offline-only experiment under genuinely congested, dynamic, randomized multi-slice traffic with real shared-PRB contention shows DQN, once trained with a correctly decaying exploration schedule, preserves URLLC SLA compliance under either reward ($22.6\% \rightarrow 30.9\%$ SLA-only, $\rightarrow 27.0\%$ QoE-aware) by trading off eMBB ($\rightarrow 7.9\%/8.2\%$) and mMTC ($\rightarrow 9.2\%/10.8\%$) – the URLLC-first outcome our prior simulated work targeted, now shown under real cross-slice scarcity, and a property of the trained policy rather than the specific reward formulation. We also report where offline-to-live transfer does and does not hold, and position this evidence against the five validation directions our review's proposed QoE-aware GAT-MARL-FL framework calls out.**

***Index Terms*—O-RAN, network slicing, deep reinforcement learning, admission control, quality of experience, URLLC prioritization, live testbed**

I. Introduction

Network slicing lets one physical O-RAN deployment serve heterogeneous service classes (eMBB, URLLC, mMTC) under per-slice SLA guarantees, but static, threshold-based control cannot adapt PRB allocation as demand shifts [4]. DRL is a natural fit: our own prior work formulated single-gNB SLA-aware admission control (SAC) [1] and extended it to a third slice with a multi-gNB load-balancing term [2], reporting ≈99.5% and 99.63/88.41/97.64% per-slice URLLC/eMBB/mMTC SLA satisfaction respectively – both validated entirely in simulation. Our systematic review of the 2024–2026 literature [3] (6,286 records screened to 67 core studies) found this pattern holds across the field and crystallized it as one of five recurring gaps – alongside QoE inference, topology awareness, privacy preservation, and scalability – motivating a proposed QoE-aware GAT-MARL-FL framework (graph-attention state encoding [5], centralized-training multi-agent control, and federated learning [6] across gNBs) to close them.

This paper makes four contributions. **(1)** We realize the SAC admission-control MDP from [1], [2] on a real OAI gNB over its native, RIC-free E2 control loop – not simulation – directly addressing the validated-deployment gap, and surface a structural finding for anyone attempting this on real O-RAN hardware: the real E2 agent exposes no per-flow accept/reject primitive, only a per-slice PRB ratio ceiling, so admission must be realized as *ceiling nudging*. **(2)** We center the evaluation on QoE, comparing a QoE-aware reward directly against the SLA-only reward of [1], [2] on the same live hardware. **(3)** We report offline-simulated and live results side by side for the same frozen policies, showing where simulation-to-real transfer holds and where it does not. **(4)** We build a genuinely congested, dynamic, randomized multi-slice scenario with real shared-PRB contention across slices and show a properly-trained DQN policy, under either reward, learns to preserve URLLC at the deliberate expense of eMBB/mMTC – the same URLLC-first objective as [1], [2], now under real cross-slice scarcity rather than isolated per-slice utility.

Scope. Single gNB, three UEs (one per slice), rfsim. The three slices already contend for one shared 106-PRB pool at this gNB – a genuine, small-scale multi-agent coordination problem. An explicit multi-agent (GAT-MARL) controller over that shared pool, multi-gNB coordination, and federated learning – our review's proposed framework in full [3] – are the journal-length follow-up's target (Section V), not attempted here.

II. System Architecture and Testbed

Fig. 1 shows the control loop: an Open5GS core with three slice-specific SMF/UPF pairs, an OAI gNB (ORANSlice fork [7], RIC-free UDP E2 agent built into the gNB binary, distinct from vanilla OAI's FlexRIC [8], [9]), and three `nr-uesoftmodem` UEs, one per slice. The E2 agent's only control primitive is `slicing_control_m`{sst, sd, min_ratio, max_ratio} [10], [11] – a per-slice PRB ratio ceiling enforced by the MAC scheduler, with no raw per-flow

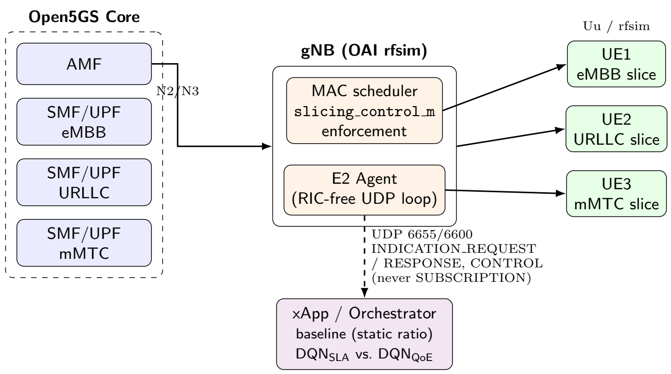

Fig. 1: Testbed control loop: Open5GS core, OAI gNB (MAC scheduler + E2 agent), 3 rfsim UEs (one per slice), and the xApp/orchestrator realizing baseline vs. DQN.

TABLE I: Testbed and campaign parameters

Parameter	Value
gNB / E2 primitive	OAI rfsim, 106 PRB, μ=1; `slicing_control_m` ratio ceiling
Slices	eMBB (4 Mbps), URLLC (300 kbps), mMTC (50 kbps bursty, 2 s/6 s)
Ceilings (eMBB/URLLC/mMTC)	12 / 4 / 3 (below each slice's measured organic demand)
QoE weights α, β, γ	1.0, 0.2, 0.5 (Eq. 2)
Arms	baseline (static), DQN$_{\text{SLA}}$, DQN$_{\text{QoE}}$
Live episodes	3 seeds $\times$ 5 ep./arm = 45 learned + 15 baseline
Offline training	300 ep. $\times$ 3 seeds/arm; frozen weights at live deployment

accept/reject hook. The binary admission decision of [1], [2] is therefore realized as ceiling nudging: accept raises a slice's `max_ratio` toward a calibrated cap, reject lowers it toward its floor. Ratio caps were calibrated by direct live measurement rather than assumed linear (Table I): the naive assumption that the 0–100 ratio scale maps linearly onto real PRBs at this cell's 106-PRB/μ=1 configuration does not hold, so each slice's cap was set empirically below its own measured organic demand. The static baseline pins each slice's ceiling-step size to zero (no accept/reject decision can move it for the whole episode); learned arms are DQN [12] policies trained offline under an SLA-only reward (Section III, Eq. 1) or a QoE-aware reward (Eq. 2) against a closed-loop synthetic KPM source, then evaluated live with frozen weights.

III. Problem Formulation

Each admission decision at step t accepts or rejects incoming demand for slice $s \in \mathcal{S} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$, realized as the ceiling adjustment above. Two reward formulations are compared. The SLA-only reward (eq. 2 of [1], [2]):

$$r_t^{\text{SLA}} = \sum_{s\in\mathcal{S}} w_s c_s n_{s,t}^{\text{acc}} \;-\; \sum_{s\in\mathcal{S}} \mathbb{1}[v_{s,t}]\, p_s \;-\; \eta\,\bar{\rho}_t\, N_t^{\text{acc}} \qquad (1)$$

where w_s/p_s are slice s's priority weight and violation penalty (URLLC highest of the three, Table I), $n_{s,t}^{\text{acc}}$ is slice s's accepted count, $\mathbb{1}[v_{s,t}]$ flags an SLA violation, and $\eta\bar{\rho}_t N_t^{\text{acc}}$ penalizes total accepted volume by mean cluster congestion $\bar{\rho}_t$. The QoE-aware reward (eq. 9 of [3]), used to center this paper's evaluation on QoE rather than SLA alone:

$$r_t^{\text{QoE}} = \alpha\,\widehat{\text{MOS}}_t \;-\; \beta\,\bar{\rho}_t N_t^{\text{acc}} \;-\; \gamma\,\overline{\text{SLA\_viol}}_t \qquad (2)$$

with α=1.0, β=0.2, γ=0.5, $\widehat{\text{MOS}}_t \in [0,1]$ the mean inferred per-slice MOS (IQX closed-form prior [13] refined by a per-slice LSTM, calibrated against ITU-T P.1203 [14] objective labels for eMBB and ACR-style task-success scoring for URLLC/mMTC), and $\overline{\text{SLA\_viol}}_t \in [0,1]$ the mean per-slice violation *severity* (continuous margin, not a flat per-violation charge). Section IV-C's congested scenario additionally enforces a genuine shared-PRB-pool constraint absent from the base offline simulator: each slice's requested PRBs $\text{req}_{s,t}$ are served in full only while the pool has slack, otherwise every slice is rationed proportionally to its own request share,

$$\text{served}_{s,t} = \begin{cases} \text{req}_{s,t}, & \sum_{s'} \text{req}_{s',t} \le B \\ \text{req}_{s,t} \cdot \dfrac{B}{\sum_{s'} \text{req}_{s',t}}, & \text{otherwise} \end{cases} \qquad (3)$$

with B the shared PRB budget. This rationing rule is priority-neutral by construction: any URLLC protection observed downstream must come from the policy's reward-driven behavior, not from the physics.

IV. Results

A. Live Testbed: DQN vs. Static Baseline

Fig. 2 shows the mechanism directly: baseline's commanded ceiling is bit-for-bit flat at its static nominal ratio for the entire episode, while DQN rides eMBB's ceiling from floor to its calibrated cap within $\sim$13 steps and jumps URLLC/mMTC to their caps immediately, holding both for the remainder. The direct consequence, across 45 live episodes (3 seeds $\times$ 5 episodes $\times$ 2 reward modes) on real hardware: DQN under both the SLA-only and QoE-aware reward holds $100.0\pm0.0\%$ SLA compliance on every slice, in every episode, against the static baseline's $73.6\pm18.6\%$ (bootstrap 95% CI $[53.4, 93.4]$), with a worst single episode of 0.0% for baseline versus 100.0% for DQN (Fig. 3, Table II).

Baseline's compliance is real but *seed-dependent* (60.3%/100.0%/60.7% across the three seeds) – the practical value demonstrated is reliability under variable conditions, not rescuing an otherwise-always-failing baseline. We do not report Cohen's d: every DQN arm's per-seed compliance has zero variance, so a pooled-variance effect size would be driven entirely by baseline's own spread. Table II's Inferred MOS column shows the QoE side of this result is more nuanced than SLA compliance alone: eMBB and URLLC MOS sit low ($\approx$ 1.2–1.5/5) and are overwhelmingly policy-independent, while mMTC sits high ($\approx$ 4.6–4.8/5), equally policy-independent – meeting the SLA compliance budget is not the same as achieving high inferred QoE for

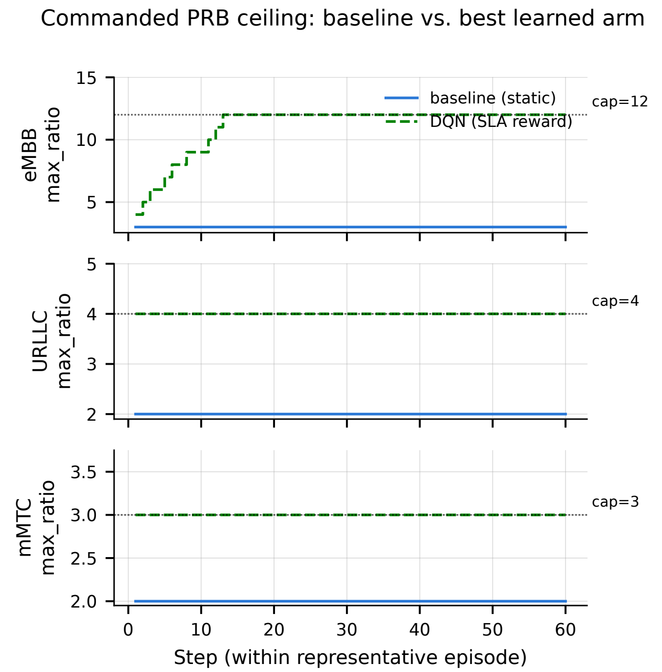

Fig. 2: Commanded PRB ceiling per slice over one representative episode: baseline (flat) vs. DQN (rides to the calibrated cap, annotated per slice).

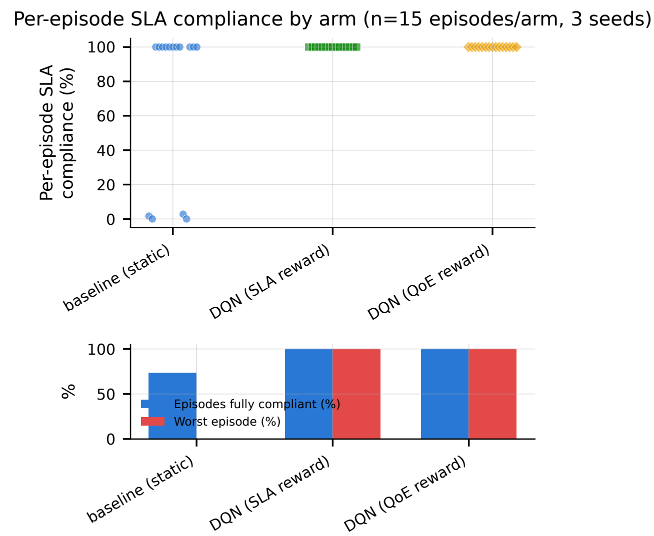

Fig. 3: Per-episode SLA compliance (%), $n = 15$ episodes/arm. Top: one point per episode – DQN sits at exactly 100.0% in all 15 episodes; baseline shows a genuine bimodal split. Bottom: fraction of episodes fully SLA-compliant and worst single-episode compliance.

TABLE II: Live-campaign results, mean ± std over 3 seeds (worst ep. pooled across all episodes, not a mean-of-means).

Arm	Slice	SLA compl. (%)	Worst ep. (%)	Inferred MOS
baseline	eMBB	73.7±18.6	0.0	1.26±0.02
	URLLC	73.4±18.8	0.0	1.24±0.35
	mMTC	73.8±18.5	0.0	4.65±0.09
DQN (SLA)	eMBB	**100.0±0.0**	**100.0**	1.22±0.00
	URLLC	**100.0±0.0**	**100.0**	1.49±0.05
	mMTC	**100.0±0.0**	**100.0**	4.78±0.00
DQN (QoE)	eMBB	**100.0±0.0**	**100.0**	1.22±0.00
	URLLC	**100.0±0.0**	**100.0**	1.55±0.06
	mMTC	**100.0±0.0**	**100.0**	4.77±0.00

these two slices on this traffic profile, exactly the gap a QoE-aware reward is meant to surface rather than hide.

B. Offline vs. Online: Sim-to-Real Transfer

DQN is trained offline against a closed-loop synthetic KPM source, then evaluated live with frozen weights; Fig. 4 compares each arm's offline-predicted compliance (final 5 rollup episodes/seed, pooled across 3 training seeds) against its live-measured compliance. The result is not a clean diagonal. eMBB transfers exactly: 100.0% $\rightarrow$ 100.0% for both reward modes. URLLC and mMTC do not: offline-predicted compliance is exactly 0.0% for both slices, for both arms, throughout all 300 training episodes, while the same frozen policies measure 100.0% live. This is not an accidental scale mismatch: the offline training config deliberately pins URLLC/mMTC's ceiling at their own nominal ratio with zero elastic headroom, an oversubscription-by-construction regime that never intended offline compliance to be a live predictor for these two slices. The consequence holds regardless of intent: the SLA-only reward had no compliance-based learning signal to differentiate policies on URLLC/mMTC during offline training – whatever separates live URLLC/mMTC behavior across arms was not something the offline reward could have selected for. We report this as this paper's most actionable near-term fix (Section V), not as a result to spin.

C. Congested, URLLC-Prioritized Admission

Sections IV-A–IV-B use one constant per-slice demand level per episode. This section asks a sharper, QoE-motivated question offline: under genuinely congested, dynamic, randomized multi-slice traffic with real shared-PRB contention (Eq. 3), does DQN preserve URLLC's SLA compliance ahead of eMBB/mMTC, consistent with URLLC already carrying the highest reward priority (Table I)? This required two fixes to the base offline simulator: matching its ceiling headroom to the live config (Section IV-B's finding), and adding the shared-pool constraint of Eq. 3 (calibrated to B=8, smaller than the three ceilings' combined maximum of 19 PRB, so maxing every ceiling simultaneously creates genuine scarcity), since the base simulator computed each slice's served PRBs independently and never let one slice's ceiling take capacity from another. Per-episode demand is additionally randomized ($0.4\times$–$1.3\times$ each slice's mean) and remains dynamic within-episode.

Table III and Fig. 5 show the held-out result, trained with a corrected exploration schedule: an earlier version of

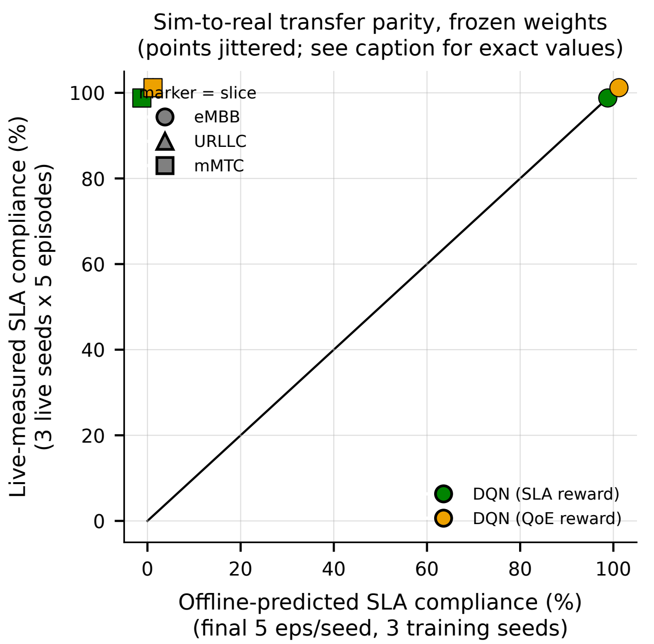

Fig. 4: Offline-predicted vs. live-measured SLA compliance, frozen weights, DQN under both reward modes (points jittered for visibility; both arms coincide exactly per slice). eMBB sits on $y{=}x$; URLLC/mMTC sit at (offline$=0$, live$=100$).

TABLE III: SLA compliance (%) under congested/dynamic/randomized traffic with genuine shared-PRB contention; held-out seeds $\times$ 15 episodes/arm, frozen weights.

Arm	URLLC	eMBB	mMTC
baseline (static)	22.6	34.5	19.0
DQN (SLA reward)	**30.9**	**7.9**	**9.2**
DQN (QoE reward)	27.0	8.2	10.8

this experiment left `DQNAdmissionPolicy`'s per-episode epsilon decay uncalled, so training-time action selection was uniform-random throughout (confirmed via checkpoint inspection, $\epsilon = 1.0$ after $\sim$18,000 gradient steps) rather than the intended decay from full exploration to near-greedy; Q-learning is off-policy and still learns from the randomly-collected, reward-labeled transitions, but this is not the schedule Table I intends, so we fixed it and retrained both arms before reporting final numbers.[1] With the fix, both DQN arms improve URLLC above baseline ($22.6\% \rightarrow 30.9\%$ SLA-only, $\rightarrow 27.0\%$ QoE-aware) by actively *sacrificing* the other two slices ($34.5\% \rightarrow 7.9\%$/8.2% eMBB, $19.0\% \rightarrow 9.2\%$/10.8% mMTC) – a genuine, learned priority tradeoff, not a policy that merely rides every ceiling upward. Critically, this is

[1]A2C [15] was also trained under this protocol (unaffected by the epsilon-schedule issue above – it has no ϵ-greedy step) and converged to a fully deterministic, state-independent policy under both reward modes regardless of exploration or entropy-regularization interventions – a genuine algorithm-dependent limitation, not a tuning problem, and omitted from the headline comparison here.

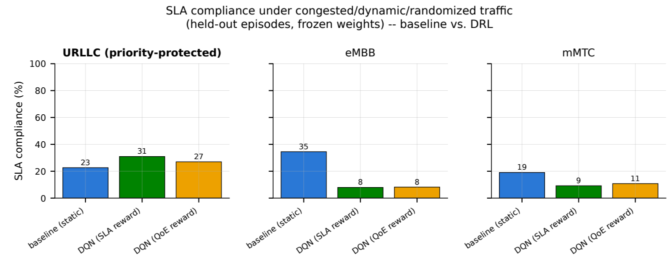

Fig. 5: SLA compliance under congested/dynamic/randomized traffic with genuine shared-PRB contention, baseline vs. DQN under both reward modes, URLLC panel bolded. Both DQN arms lead on URLLC and visibly sacrifice eMBB/mMTC to do it – not a QoE-reward-specific effect.

no longer reward-mode-specific: an earlier, since-corrected run (uniform-random training-time exploration) had shown only the QoE-shaped reward producing this tradeoff, with the SLA-only reward instead improving eMBB without sacrificing it; with exploration correctly decaying, `DQN(SLA)` shows the same tradeoff as `DQN(QoE)`, and if anything a marginally larger URLLC gain. We read this as evidence that the URLLC-priority tradeoff is a property of a properly-trained DQN policy under this reward priority structure (Table I's priority_weight/violation_penalty), not a QoE-reward-specific effect – a more robust but less differentiated finding than we initially reported. This is the same URLLC-first objective as [1], [2], now shown under a materially harder condition than either prior study modelled – a real constraint that the three slices' served PRBs sum to a shared, insufficient budget, rather than each slice's utility optimised in isolation. This offline, preliminary result is not yet live-validated (Section VI); reproducing genuine cross-slice scarcity on the real rig is the clearest next experimental step.

V. Discussion

A. What This Paper Achieves Relative to Our Review

Our review [3] identified five recurring gaps – validated deployment, QoE inference, topology awareness, privacy preservation, and scalability – and proposed a QoE-aware GAT-MARL-FL framework conditioned on five validation directions. This paper **closes** the validated-deployment gap for the SAC formulation: Sections IV-A–IV-B show DRL-based admission control, realized as ceiling nudging on real E2 hardware, reliably beats a static ratio under constant load. It contributes **partial** evidence toward QoE inference (a calibrated IQX+LSTM mapper exists, but Section IV-A shows MOS is largely policy-independent on this traffic profile – calibration exists, discriminative power does not yet) and toward topology/scalability (Section IV-C is a first, offline demonstration that shared-resource contention across slices is learnable by a single-agent controller, not yet tested across multiple gNBs or with a topology-aware state). Privacy preservation (FL) remains **untouched**: this is a single-rig, single-tenant testbed by construction. Table IV states this plainly.

TABLE IV: This paper against [3]'s five gaps.

Gap	Status here
Validated deployment	**Closed** for SAC: real E2 loop, real gNB (Sec. IV-A–IV-B).
QoE inference	**Partial**: mapper calibrated, but MOS largely policy-independent (Sec. IV-A).
Topology / scalability	**Partial**: single gNB; Sec. IV-C is a first offline contention result.
Privacy (FL)	**Untouched**: no federated learning; single-rig.
Resilience / scarcity	**Partial**: rig recovery demonstrated; Sec. IV-C's scarcity is offline-only.

B. What's Next: Toward the Journal-Length Follow-Up

Three items follow directly from what this paper establishes. **(1)** Recalibrate the offline environment's URLLC/mMTC demand and backlog scaling against live measurement, using the same methodology already applied to eMBB (Section IV-B) – the paper's most actionable near-term fix, and a precondition for training any new checkpoint under a different architecture. **(2)** Live-validate Section IV-C's shared-pool-contention finding, which needs genuine combined-demand scarcity across all three UEs simultaneously, not per-slice oversubscription alone. **(3)** Replace the single-agent-per-slice arms with the GAT-encoded, centralized-training-decentralized-execution multi-agent controller our review proposes [5], motivated directly by Section IV-C's finding that a single-agent policy learns the URLLC-priority tradeoff only under careful, manual reward shaping; extend to multiple gNBs before layering on the federated-learning component [6]. None of these three is attempted here; each is scoped by what Sections IV-A–IV-C already establish, and is the direct target of our #5 journal-length follow-up.

VI. Limitations

Results are from OAI's rfsim radio simulator, not over-the-air deployment. The testbed runs on a single shared desktop machine and was subject to an intermittent radio-link failure under memory pressure, mitigated by an automatic health-check-and-restart between episodes (verified to recover in every observed instance); every evaluation episode counted here completed and was log-verified end to end. A single gNB and three UEs were used, so paper #2's multi-gNB load-balancing term is not evaluated here. Live evaluation used 3 seeds per arm. Per-slice MOS is an inferred quantity, not a measurement from real end-user subjective scoring. Section IV-C's result is offline-only, evaluates DQN only (A2C's contrasting, degenerate behavior is reported in a footnote there but not part of this paper's live-validated results), and has not been reproduced on the live rig.

VII. Conclusion

This paper realizes the SAC admission-control MDP from [1], [2] on a real OAI gNB over its native RIC-free E2 control loop, closing the validated-deployment gap our own systematic review [3] identified, and centers the evaluation on QoE rather than SLA compliance alone. Across 45 live episodes on real hardware, DQN holds $100.0 \pm 0.0\%$ SLA compliance under both SLA-only and QoE-aware rewards against a static baseline's seed-dependent $73.6 \pm 18.6\%$ (worst episode 0.0%). Comparing offline-simulated and live results side by side shows transfer holds exactly for eMBB but not for URLLC/mMTC, a diagnosed, fixable limitation of the offline training environment rather than a robustness result. A preliminary, offline-only follow-up under genuine cross-slice PRB contention shows a properly-trained DQN policy learns to preserve URLLC SLA compliance under either reward ($22.6\% \rightarrow 30.9\%$ SLA-only, $\rightarrow 27.0\%$ QoE-aware) by actively trading off eMBB ($\rightarrow$ 7.9%/8.2%) and mMTC ($\rightarrow$ 9.2%/10.8%) – the same URLLC-first objective as our prior simulated work [1], [2], now shown under real contention and, once a training-time exploration-schedule bug we found and fixed was corrected, a property of the trained policy rather than of the QoE-shaped reward specifically. The proposed GAT-MARL-FL framework of [3] remains the target of the journal-length follow-up, motivated directly by this paper's finding that single-agent control achieves the URLLC-priority tradeoff only under careful, manual reward shaping.

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